Supplemental Online Material - Tables of Measurements

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Measurement of the Nucleon F_2^n/F_2^p Structure Function Ratio by the Jefferson Lab MARATHON Tritium/Helium-3 Deep Inelastic Scattering Experiment

x	Q^2 $(\mathrm{GeV}/c)^2$	W GeV/c^2	σ_d/σ_p	Δ_{stat}	Δ_{ptp}	Δ_{syst}	Δ_{tot}
0.195	2.73	3.49	1.725	0.005	0.010	0.009	0.015
0.225	3.15	3.42	1.697	0.005	0.008	0.009	0.014
0.255	3.57	3.36	1.674	0.007	0.008	0.009	0.014
0.285	3.99	3.30	1.656	0.008	0.010	0.009	0.016
0.315	4.41	3.24	1.629	0.008	0.010	0.009	0.016
0.345	4.83	3.17	1.588	0.009	0.010	0.009	0.016
0.375	5.25	3.10	1.544	0.013	0.016	0.008	0.023

Table 1: The ratio of deuteron to proton DIS cross section at the selected x, Q^2, and W kinematics of MARATHON. Listed are the statistical (stat), point-to-point (ptp), and overall/scale systematic (syst) components of the total (tot) error. The latter is the quadrature sum of the three components. The overall/scale systematic component of $\pm 0.55\%$ is due to the uncertainties in the nominal gas target densities of the hydrogen and deuterium gases (combined in quadrature).

x	Q^2 $(\mathrm{GeV}/c)^2$	W GeV/c^2	σ_h/σ_t	Δ_{stat}	Δ_{ptp}	Δ_{syst}	Δ_{tot}
0.195	2.73	3.49	1.112	0.003	0.005	0.008	0.010
0.225	3.15	3.42	1.124	0.003	0.004	0.008	0.010
0.255	3.57	3.36	1.141	0.004	0.004	0.008	0.010
0.285	3.99	3.30	1.160	0.005	0.005	0.008	0.011
0.315	4.41	3.24	1.154	0.005	0.004	0.008	0.011
0.345	4.83	3.17	1.171	0.006	0.005	0.008	0.011
0.375	5.25	3.10	1.177	0.008	0.006	0.008	0.013
0.405	5.67	3.03	1.219	0.009	0.004	0.009	0.014
0.435	6.09	2.96	1.206	0.010	0.005	0.009	0.014
0.465	6.51	2.89	1.226	0.010	0.004	0.009	0.014
0.495	6.93	2.82	1.235	0.010	0.004	0.009	0.014
0.525	7.35	2.74	1.260	0.011	0.004	0.009	0.015
0.555	7.77	2.67	1.267	0.011	0.004	0.009	0.015
0.585	8.19	2.59	1.268	0.012	0.004	0.009	0.016
0.615	8.61	2.50	1.268	0.013	0.004	0.009	0.016
0.645	9.03	2.42	1.292	0.013	0.004	0.009	0.017
0.675	9.45	2.33	1.289	0.014	0.004	0.009	0.018
0.705	9.87	2.24	1.309	0.014	0.004	0.009	0.017
0.735	10.3	2.14	1.302	0.013	0.004	0.009	0.017
0.765	10.7	2.04	1.316	0.014	0.004	0.009	0.017
0.795	11.1	1.94	1.312	0.015	0.004	0.009	0.018
0.825	11.9	1.84	1.301	0.017	0.004	0.009	0.020

Table 2: The helion to triton DIS cross section ratio (after normalization, see text) at the x, Q^2, and W kinematics of MARATHON. At the highest central-x kinematics of 0.825 only electron events with $W > 1.74~\mathrm{GeV}/c^2$ were included in the determination of the $A = 3$ yields. Listed also are the statistical (stat), point-to-point (ptp) and overall/scale systematic (syst) components of the total (tot) error. The latter is the quadrature of the three components.

x	$\mathcal{R}_{ht}$	$\Delta\mathcal{R}_{ht}$	F_2^n/F_2^p	Δ_{stat}	Δ_{ptp}	Δ_{syst}	Δ_{tot}
0.195	0.9989	0.0009	0.724	0.005	0.011	0.016	0.020
0.225	0.9990	0.0009	0.701	0.006	0.008	0.016	0.019
0.255	0.9991	0.0009	0.668	0.008	0.008	0.015	0.019
0.285	0.9993	0.0008	0.635	0.008	0.009	0.014	0.019
0.315	0.9997	0.0009	0.647	0.010	0.008	0.015	0.019
0.345	1.0003	0.0008	0.618	0.010	0.008	0.014	0.019
0.375	1.0010	0.0008	0.610	0.013	0.010	0.014	0.021
0.405	1.0019	0.0008	0.547	0.014	0.006	0.013	0.020
0.435	1.0029	0.0007	0.567	0.015	0.007	0.013	0.021
0.465	1.0039	0.0007	0.540	0.015	0.006	0.013	0.020
0.495	1.0049	0.0007	0.528	0.014	0.006	0.012	0.020
0.525	1.0058	0.0007	0.496	0.015	0.006	0.012	0.020
0.555	1.0067	0.0007	0.489	0.015	0.006	0.012	0.020
0.585	1.0074	0.0008	0.489	0.016	0.006	0.012	0.020
0.615	1.0081	0.0009	0.489	0.016	0.005	0.012	0.021
0.645	1.0087	0.0010	0.461	0.016	0.006	0.011	0.020
0.675	1.0093	0.0013	0.466	0.018	0.006	0.011	0.022
0.705	1.0098	0.0017	0.442	0.016	0.005	0.011	0.020
0.735	1.0104	0.0020	0.451	0.016	0.005	0.011	0.020
0.765	1.0111	0.0024	0.436	0.016	0.006	0.011	0.020
0.795	1.0118	0.0030	0.441	0.017	0.006	0.011	0.022
0.825	1.0125	0.0043	0.455	0.020	0.009	0.011	0.024

Table 3: The F_2^n/F_2^p ratio for the MARATHON x kinematics. Listed also are the ratio's statistical (stat), point-to-point (ptp), and overall/scale (syst) components of the total (tot) error. The latter is the quadrature of the three components. Also listed are the values for the $\mathcal{R}_{ht}$ super-ratio and their uncertainties used in the F_2^2/F_2^p ratio extraction (see text).